STATISTICAL ANALYSIS PLAN

TROPIC / EFC6193 - Industry-Grade Clinical Programming Baseline

Cabazitaxel + Prednisone vs Mitoxantrone + Prednisone in Metastatic Castration-Resistant Prostate Cancer

Document attribute	Value
Version	4.0 controlled draft
Date	2026-06-25
Status	Sponsor-approvable SAP draft for remediation and programming control; not sponsor-approved until signed.
Study	EFC6193 / TROPIC / NCT00417079
Primary data limitation	Current repo contains real MP patient-level data and reconstructed/synthetic CbzP data. Full two-arm confirmatory inference requires complete authoritative two-arm patient-level data.
Document purpose	Define the statistical and programming authority needed to rebuild the pipeline as a serious 2026 clinical programming work product.

Control statement: This SAP is intentionally stricter than the current repo. Programs, TFLs, Define-XML, reviewer guides and eCTD outputs must conform to this SAP and its release gates. Generated artifacts cannot retroactively establish prespecification.

Approval role	Name / signature	Date	Decision
Lead Statistician			Approve / Revise / Reject
Lead Statistical Programmer			Approve / Revise / Reject
Independent QC Programmer			Approve / Revise / Reject
Clinical Domain Lead			Approve / Revise / Reject
Data Standards Lead			Approve / Revise / Reject
Quality / Validation Representative			Approve / Revise / Reject

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1. Administrative control and document status

This document is the controlling Statistical Analysis Plan (SAP) draft for rebuilding the TROPIC clinical programming pipeline as a serious regulated-workflow artifact. It is not the original Sanofi SAP and does not assert sponsor approval. It is designed to be approvable after sponsor/statistical review, source-data reconciliation, and closure of release gates.

Control item	Requirement
Change control	Any change after approval requires a versioned SAP amendment, rationale, affected outputs, affected ADaM/Define metadata, and signer approval.
Traceability	Every analysis result must trace to SAP section, TFL shell, ADaM dataset/variable(s), SDTM/source predecessor(s), program, QC program and run manifest.
Evidence standard	No claim is accepted without a file, page/section, dataset, program or validator report reference.
Submission readiness	The current repository is not submission-ready until audit Critical findings F-001 through F-005 are closed and Major findings are either closed or formally risk-accepted.
Portfolio constraint	The repository may remain public/portfolio-oriented, but clinical claims must be controlled as non-confirmatory unless supported by authoritative complete data and approval.

1.1 Roles and responsibilities

Role	Accountability
Lead Statistician	Owns SAP scientific content, estimands, multiplicity, statistical models, interpretation and final approval.
Lead Statistical Programmer	Owns production programming implementation, ADaM/TFL standards alignment, run manifest and programmer sign-off.
Independent QC Programmer	Owns independent derivation and result reproduction, discrepancy triage and QC sign-off.
Clinical Domain Lead	Confirms endpoint clinical meaning, event adjudication assumptions, safety groupings and medically important caveats.
Data Standards Lead	Owns CDISC conformance, Define-XML/ARM, controlled terminology and metadata/data concordance.
Quality / Validation Representative	Owns process validation evidence, Part 11 control assessment, audit-trail retention and release readiness.

1.2 Document deliverable set controlled by this SAP

- ADaM production datasets and transport files.
- Independent R/pharmaverse validation datasets and reconciliation outputs.
- Tables, figures and listings in the controlled TFL catalog.
- Define-XML v2.1 with ARM and value-level metadata.
- ADRG, SDRG, traceability matrix, conformance reports and eCTD study-data package.
- Run manifest tying source data, programs, logs, outputs, validator results and hashes to one reproducible execution.

2. Evidence authority and source hierarchy

The SAP resolves conflicts using an explicit source hierarchy. Protocol and publication evidence are clinical/statistical sources; generated code and documents are implementation evidence only.

Rank	Source authority	Governance use
1	Protocol Amendment 5, 21-Jul-2008	Trial objectives, design, populations, endpoint concepts, randomization strata, treatment schedules, sample size assumptions and planned primary analysis.
2	Corrected de Bono et al. Lancet 2010 publication	Public final-analysis cut-off, published ITT/safety counts, efficacy/safety results, and trial-era terminology confirmation.
3	Patient-level source data made available through PDS/Sanofi	Actual data source for repo-executable real MP analyses and source timing limitations.

Rank	Source authority	Governance use
4	CRF and source define metadata	Collection context and source-domain interpretation.
5	This SAP v4.0 after approval	Programming and analysis control for this project.
6	ADaM spec, Define-XML, reviewer guides, programs and TFLs	Implementation artifacts; must conform to approved SAP and source evidence.

Evidence source: Protocol PDF pages 3, 10-12; Lancet PDF pages 1, 4-5, 8; audit/AUDIT_REPORT.md; official FDA/CDISC/ICH sources listed in References.

3. Study overview

Item	Specification
Study identifier	EFC6193 / TROPIC / NCT00417079
Design	Phase III, randomized, open-label, multi-center, multinational trial.
Indication	Metastatic castration-resistant prostate cancer after docetaxel-containing therapy.
Treatment arms	MP: mitoxantrone 12 mg/m^2 IV Day 1 every 21 days plus prednisone 10 mg daily. CbzP: cabazitaxel 25 mg/m^2 IV Day 1 every 21 days plus prednisone 10 mg daily.
Randomization	1:1 IVRS randomization stratified by disease measurability per RECIST and ECOG performance status (0-1 versus 2).
ITT randomized	Published ITT: 755 total, 377 MP and 378 CbzP.
Safety treated	Published safety: 371 per arm.
Final public cut-off	25-Sep-2009.
Primary endpoint	Overall survival.
Secondary endpoints	PFS, PSA response/progression, ORR, pain response/progression, safety and tolerability.

3.1 Treatment regimens and key administration rules

Arm	Regimen	Operational programming notes
MP	Mitoxantrone 12 mg/m^2 IV over 15-30 minutes on Day 1 of each 21-day cycle plus prednisone 10 mg orally daily.	Maximum 10 cycles. Dose reductions and delays summarized from EX/SUPPEX. Safety summaries use actual treatment received.
CbzP	Cabazitaxel 25 mg/m^2 IV over 1 hour on Day 1 of each 21-day cycle plus prednisone 10 mg orally daily.	Premedication with antihistamine, corticosteroid and H2 antagonist. In current repo, CbzP patient-level records are reconstructed/synthetic unless complete source IPD is supplied.

3.2 Assessment schedule

Assessment area	Schedule / source basis	Analysis implication
Survival	Follow-up through final analysis cut-off and last known alive/death information.	OS censoring and event status must retain source date and cut-off logic.
Tumour imaging	Baseline, end of even-numbered treatment cycles, suspected progression and end of treatment.	RECIST v1.0 response and TTUMOR/PFS component dates.
PSA	Baseline and repeated on study per protocol schedule.	Response requires confirmation; progression uses nadir-based rule by responder status.
Pain	Daily PPI and analgesic use over 7 days before visit.	Visit evaluability requires at least 5 of 7 daily values.
Haematology/labs	CBC on Cycle Day 1, Day 8 and Day 15 with protocol safety monitoring.	ADLB visit windows must include CxD1 pre-dose and nadir/recovery windows.
AE/safety	AE collection and grading through treatment and safety follow-up.	TEAE window and CTCAE v3.0 grading drive ADAE/ADLB outputs.

Evidence source: Protocol synopsis and statistical considerations; de Bono et al. Lancet 2010 summary and methods.

4. Data provenance and analysis layers

This is the central integrity rule for the project: real MP patient-level analyses and reconstructed CbzP demonstration analyses must never be blurred. The SAP permits both layers, but assigns different evidentiary status.

Layer	Data basis	Permitted claims	Programming controls
Trial target layer	Original randomized TROPIC design and published results.	Defines intended clinical questions and validation targets.	Use for objective, endpoint and validation-target specification.
Repo real-data layer	Real MP patient-level SDTM/ADaM available in repo/PDS source.	Permits real single-arm MP summaries and validation of MP-side derivations.	SAS/R reconciliation must compare MP records without lossy normalization.
Synthetic comparator layer	CbzP reconstructed from published KM curves/at-risk tables, PH scaling and fixed-seed marginal sampling.	Permits demonstration of pipeline mechanics, shells, figures and exploratory methods only.	Every output must label synthetic comparator and non-confirmatory status.
Future submission layer	Complete authoritative two-arm IPD, approved SAP, clean CDISC package and validated environment.	Permits submission-facing clinical analysis only after gates pass.	Requires full rerun, P21/CORE/Define validation, eCTD integrity and Part 11/process evidence.

5. Analysis populations

Population	Definition	Treatment assignment	Use	Target / current status
Intent-to-treat (ITT)	All randomized patients.	Randomized treatment.	Primary efficacy and most secondary efficacy analyses.	Published N=755. Current repo combined N=749 is not the original ITT and must not be labeled trial ITT.
Safety / all-treated	All patients receiving at least part of one dose.	Treatment actually received.	Safety, exposure, dose modification, deaths within safety window.	Published N=371 per arm. Current real layer has MP N=371.
Per-protocol	ITT patients without major protocol deviations.	Randomized treatment unless deviation requires exclusion.	Sensitivity/supportive only.	Must be source-flagged or derivable from protocol deviations.
PSA response evaluable	Baseline PSA >=20 ug/L plus evaluable confirmatory PSA assessments.	Randomized treatment.	PSA response rate.	Published 325 MP, 329 CbzP; programming must not use all PSARESP records.
Measurable disease	Baseline measurable lesion per RECIST v1.0.	Randomized treatment.	ORR and tumour-response summaries.	Published 204 MP, 201 CbzP.
Pain response evaluable	Baseline median PPI >=2 or mean analgesic score >=10 with diary evaluability.	Randomized treatment.	Pain response.	Published 168 MP, 174 CbzP.

6. General statistical principles

6.1 Analysis conventions

- All efficacy tests are two-sided unless explicitly stated otherwise.
- Time-to-event analyses use days as the stored analysis scale. Months are reported as days divided by 30.4375 unless a shell specifies otherwise.
- Confidence intervals are two-sided 95% intervals unless the output shell specifies another interval.
- Counts and percentages use the relevant analysis population denominator. Percentages are displayed to one decimal place unless counts are small or shell-specific precision is required.
- Treatment labels are MP and CbzP. Current repo outputs must annotate whether CbzP is reconstructed/synthetic.
- No data-driven changes to endpoint definitions, windows, populations or estimators are permitted after SAP approval without amendment.

6.2 Baseline, visits and windows

- Baseline is the last non-missing value on or before first dose unless the endpoint definition requires randomization-date baseline.
- Post-baseline excludes baseline and pre-dose values unless explicitly part of a Cycle Day 1 pre-dose laboratory window.

- CBC windows include scheduled Cycle Day 1 pre-dose, Day 8 and Day 15 assessments.
- When multiple values fall in a window, select one analysis record by minimum absolute distance to target day; ties use worst toxicity grade, then latest available record, unless a parameter-specific rule overrides this.
- Unscheduled assessments may be used for event detection if clinically valid and source-date reliability is adequate.

6.3 Software and reproducibility

- Production and QC are dual-language where feasible: SAS production and R/pharmaverse validation.
- Each run must bind Git commit, program versions, inputs, outputs, logs, hashes, validator versions and reviewer disposition in a run manifest.
- Random generation for synthetic comparator data requires deterministic seed capture, seed rationale and hash-bound output.
- SAS and R logs must be reviewed and fail on unapproved errors, warnings, invalid INPUT, uninitialized variables and problematic merge notes.

6.4 Statistical model and display standards

Analysis type	Primary method	Display requirement	QC requirement
Time-to-event	Kaplan-Meier, stratified log-rank, stratified Cox PH where applicable.	Median, 95% CI, events/N, censoring count, HR, CI, p-value and at-risk table for figures.	Independent reproduction of event/censor counts, median, HR and p-value.
Binary response	Exact/binomial CI and Fisher or CMH test if stratified comparison is justified.	n/N, percent, 95% CI, difference/ratio if shell-approved.	Population denominator and confirmation records independently verified.
Continuous/lab	Descriptive statistics and shift tables; no inferential testing unless prespecified.	N, mean, SD, median, Q1, Q3, min, max; baseline/worst shifts.	Source units, grade derivation and selected analysis record independently verified.
Safety incidence	Subject-level incidence by treatment, SOC/PT and grade/seriousness/action.	n, %, denominator footnote, coding dictionary and grade version.	Worst-grade and one-subject-once-per-category logic independently verified.
Exploratory exposure-response	Descriptive regression/plots with clearly labeled non-confirmatory status.	Effect estimates only if model assumptions and synthetic status are disclosed.	Seed, derivation, model code and synthetic flags independently checked.

7. Estimand framework

The estimands below follow ICH E9(R1): treatment condition, population, endpoint variable, intercurrent-event handling, population-level summary and estimator are specified before analysis. Because the current repo lacks complete real CbzP IPD, confirmatory interpretation is blocked even where estimand structure is complete.

Endpoint	Population	Variable	Intercurrent-event strategy	Summary / estimator	Current evidentiary status
OS	ITT	Time from randomization to death from any cause.	Treatment-policy for discontinuation/subsequent therapy; censor alive subjects at earliest last-known-alive or cut-off.	KM medians; stratified log-rank; stratified Cox HR.	Protocol-confirmed target; current two-arm repo result non-confirmatory due synthetic CbzP.
PFS	ITT	Time to first valid PSA, tumour, pain progression with disease evidence, or death.	New anti-cancer therapy before progression censors at NACTDT-1; no post-baseline assessment censors at randomization.	KM medians; stratified log-rank; stratified Cox HR.	Target estimand; component provenance must be repaired/validated.
PSA response	Baseline PSA >=20 ug/L	Confirmed >=50% PSA decline from baseline with repeat >=3 weeks.	Missing confirmation means non-response unless death/progression handling is shell-specified.	Response rate with exact CI; Fisher or CMH support.	Current programming must correct eligible denominator.
ORR	Measurable disease	Confirmed CR/PR per RECIST v1.0.	Missing/NE response not responder; new lesion/progression as PD.	Response rate with exact CI; supportive duration if available.	RECIST v1.0 required; exploratory hybrids must be labeled.
Pain response	Pain response evaluable	PPI or analgesic-score improvement maintained >=3 weeks.	Insufficient diary data yields not evaluable; missing confirmation means non-response.	Response rate and exact CI.	Requires 5-of-7 diary evaluability implementation.
Safety	Safety/all-treated	TEAEs, SAEs, grade >=3 events, deaths, lab toxicities.	Treatment discontinuation is part of safety experience; events through defined window included.	Counts/percentages; exposure-adjusted summaries if specified.	Current lab shift and synthetic-comparator safety outputs require fixes/labeling.

8. Multiplicity and Type I error control

For the original intended two-arm trial analysis, hierarchical testing is specified for the principal efficacy family. For the current repository implementation, p-values involving reconstructed CbzP are descriptive only and do not control Type I error for clinical claims.

Step	Endpoint	Population	Decision rule
1	Overall survival	ITT	Primary endpoint; proceed only if OS significant under approved alpha strategy.
2	Progression-free survival	ITT	Test only if Step 1 is met.
3	PSA response	Baseline PSA >=20 ug/L	Test only if Step 2 is met.
4	Objective response rate	Measurable disease	Test only if Step 3 is met.
Exploratory	Pain, TTPSA, TTUMOR, subgroups, exposure-response, Optimus	Endpoint-specific	No alpha protection; descriptive and hypothesis-generating.

9. Primary efficacy analysis - overall survival

9.1 Definition and censoring

- OS is time from randomization date to death from any cause.
- Subjects alive at analysis cut-off are censored at the earlier of last date known alive and 25-Sep-2009.
- Deaths after documented withdrawal but before cut-off are events if source evidence supports death date.
- Analysis variable: ADTTE where PARAMCD='OS'; CNSR=0 event, CNSR=1 censored; AVAL is days from randomization plus one if day-count convention requires inclusive duration.

9.2 Primary estimator

- KM curves by randomized treatment.
- Median OS with 95% CI by Brookmeyer-Crowley method.
- Stratified log-rank test using randomization strata: disease measurability and ECOG PS.
- Stratified Cox proportional hazards model using the same strata; report HR for CbzP versus MP with 95% CI.
- Number at risk displayed at clinically relevant months matching the publication where feasible.

9.3 Sensitivity and diagnostics

- Unstratified Cox/log-rank sensitivity.
- RMST at 12 months and optionally 18 months when follow-up supports interpretation.
- PH diagnostics using log-minus-log plots and Schoenfeld residuals where real two-arm IPD is available.
- Censoring distribution by arm and reason.

Published validation target	Value
Median OS CbzP	15.1 months (95% CI 14.1-16.3)
Median OS MP	12.7 months (95% CI 11.6-13.7)
HR CbzP vs MP	0.70 (95% CI 0.59-0.83), p<0.0001
Deaths at cut-off	513 ITT deaths, 234 CbzP and 279 MP

10. Secondary efficacy analyses

10.1 Progression-free survival

- PFS is time from randomization to the earliest of PSA progression, tumour progression by RECIST, pain progression with supporting disease evidence, or death from any cause.
- The component event date is the earliest valid date among components. Component source, date and priority must be retained in ADTTE traceability variables.
- If no post-baseline assessment exists, censor at randomization date.
- If new anti-cancer therapy begins before documented progression, censor at the day before new therapy start.
- Primary estimator mirrors OS: KM, stratified log-rank, stratified Cox HR and medians.

10.2 PSA response and PSA progression

- PSA response applies only to subjects with baseline PSA >=20 ug/L.
- Responder is confirmed >=50% decline from baseline, with repeat PSA at least 3 weeks later.
- PSA progression for non-responders is >=25% increase over nadir and absolute increase >=5 ug/L, confirmed per protocol timing.
- PSA progression for responders is >=50% increase over nadir; do not apply the >=5 ug/L absolute floor to responders unless a later approved amendment states otherwise.
- Time to PSA progression is analyzed using time-to-event methods with ITT denominator for the original trial target.

10.3 Tumour response and time to tumour progression

- RECIST v1.0 governs tumour response for subjects with measurable disease.
- ORR is confirmed CR or PR. Confirmation requires a subsequent qualifying assessment at least 4 weeks later.
- SD requires sufficient duration from study entry per RECIST v1.0 conventions.
- New lesion or unequivocal progression yields PD.
- Time to tumour progression is a time-to-event endpoint distinct from composite PFS.

10.4 Pain endpoints

- Pain instruments are McGill-Melzack present pain intensity (PPI) and analgesic score normalized to morphine equivalents.
- Visit-level PPI/analgesic summaries require at least 5 of 7 expected daily values.
- Pain response applies to subjects with baseline median PPI >=2 or mean analgesic score >=10.
- Pain response requires either >=2 point PPI reduction without analgesic-score increase, or >50% analgesic-use decrease without pain increase, maintained >=3 weeks.
- Pain progression uses increase from baseline/reference value, not nadir, and includes palliative radiotherapy for pain.

10.5 Secondary endpoint model specification matrix

Endpoint	Primary analysis set	Primary estimator	Key sensitivity
PFS	ITT	Stratified Cox/log-rank and KM medians.	Exclude assessments on/after new anti-cancer therapy; component-specific audit.
TTPSA	ITT	TTE methods using PSA progression event.	Responder-status rule audit; confirmation-window sensitivity.
TTUMOR	Measurable disease / ITT target per shell	TTE methods using RECIST PD/new lesion.	Assessment-window and missing imaging sensitivity.
TTPAIN	ITT with diary evaluability	TTE methods using pain progression event.	5-of-7 diary rule and palliative-RT-only sensitivity.
PSA response	Baseline PSA >=20 ug/L	Confirmed response rate and exact CI.	Treat missing confirmation as non-response; evaluable-only support.
ORR	Measurable disease	Confirmed CR/PR rate and exact CI.	Unconfirmed response support and NE handling display.
Pain response	Pain response evaluable	Response rate and exact CI.	Diary completeness and baseline pain subgroup displays.

11. Safety analyses

11.1 Safety population and windows

- Safety analyses use the all-treated population and actual treatment received.
- TEAE window begins at first dose and ends 30 days after last dose unless a protocol-specific SAE collection rule requires a broader window.
- AE severity uses NCI CTCAE v3.0; coding uses MedDRA v12.0 for publication reconciliation.
- The safety denominator is 371 per arm for publication targets. Current repo real patient-level safety analyses are MP-only unless complete CbzP IPD is obtained.

11.2 Adverse events

- Summaries include any TEAE, serious TEAE, grade >=3 TEAE, TEAE leading to dose reduction, delay, discontinuation or death.
- Subject-level incidence counts each subject once per SOC/PT/grade category using worst severity.
- Haematological safety events are derived from laboratory grades where the publication defines them as lab-based.

- OCCDS episode merging may be used for event episodes, but subject-level incidence must remain reviewable and traceable.

11.3 Laboratory analyses

- Laboratory analyses include baseline, scheduled cycle values, worst post-baseline grade, shift tables and nadir/recovery summaries.
- Lab shift programming must deduplicate to one baseline and one worst post-baseline record per subject/parameter before cross-tabulation.
- CTCAE grades must be traceable to source grade or independently derivable from lab value, unit and reference range.
- Key haematology parameters include neutrophils/ANC, leukocytes, hemoglobin/anaemia and platelets/thrombocytopenia.

11.4 Deaths and discontinuations

- Deaths are summarized for ITT survival and separately for safety deaths within 30 days of last dose.
- Discontinuation reasons must be produced from source/ADaM records. Placeholder listings are prohibited.
- Any death or discontinuation listing must include USUBJID, treatment, relevant dates, reason/cause, source domain and traceability variables.

12. Exposure, dose modification and Project Optimus exploratory analyses

- Exposure summaries include cycles received, treatment duration, cumulative dose, planned dose, actual dose, relative dose intensity, dose reductions and delays.
- RDI is actual cumulative dose divided by planned cumulative dose times 100, validated against SUPPEX anchors when available.
- G-CSF use is summarized by timing, cycle and prophylactic/therapeutic status where source data permits.
- Project Optimus analyses are retrospective and exploratory. They may structure dose/exposure-response thinking but cannot establish optimized dose for a historic completed trial.
- Exploratory exposure-response outputs must separate real MP-only evidence from reconstructed CbzP demonstration data.

13. Subgroup, sensitivity and supplementary analyses

- Subgroup analyses are descriptive; no multiplicity adjustment is applied.
- OS subgroup forest plots reproduce publication-defined subgroup categories where available.
- Subgroups include ECOG PS, measurable disease status, pain baseline status, PSA status, prior chemotherapy regimens, age group and other publication-defined factors.
- Subgroup counts may not sum to 755 when the publication documents missing/unknown classifications; missing status must be displayed or footnoted.
- Sensitivity analyses must be specified before execution and must not replace the primary estimator unless an amendment is approved.

14. Data handling, missing data, partial dates and visit windows

Topic	Rule
Missing efficacy outcome	Use endpoint-specific censoring or non-responder rules. Do not impute response success.
Partial dates	Do not invent date components in source data. Use conservative bounds for classification and sensitivity analyses where partial dates affect treatment-emergent or time-to-event status.
Week-offset dates	AE/DS timing stored as integer weeks is reconstructed using RFSTDTC plus (week-1)*7, with +/-3.5 day uncertainty disclosed.
Duplicate records	Define parameter-specific tie-breakers before analysis. Retain flags showing selected records.
Outliers	Do not exclude clinical values solely because they are extreme. Flag, review and document source/data-query status.
Synthetic data	Synthetic records must carry derivation/provenance flags and must not be mixed into real-data validation denominators.
Analysis-day convention	ADY is date minus reference start date plus one for dates on/after reference start; negative days follow CDISC convention.

15. CDISC implementation and metadata traceability

Dataset	Class	Structure	Variable count	Purpose
ADSL	SUBJECT LEVEL ANALYSIS DATASET	One record per subject	42	Subject-level characteristics, treatment allocations, survival, and baseline covariates.
ADEX	BASIC DATA STRUCTURE	One record per subject per parameter per visit	14	Cycle-level actual dose performance, cumulative exposure, actual/planned dose intensities, and RDI calculations.
ADCM	OCCURRENCE DATA STRUCTURE	One record per subject per medication	15	Concomitant medication usage including G-CSF prophylaxis markers, prednisone compliance, and NACT timelines.
ADAE	OCCURRENCE DATA STRUCTURE	One record per subject per adverse event	29	Treatment-emergent adverse events, severity classifications, serious AEs, and custom myelosuppressive episode merging.
ADLB	BASIC DATA STRUCTURE	One record per subject per parameter per visit	27	Longitudinal absolute laboratory measurements, CTCAE grading, Worst-in-window flags (ANL01FL), and continuous nadir models.
ADRS	BASIC DATA STRUCTURE	One record per subject per parameter per visit	13	RECIST v1.0 soft-tissue assessments, bone-lesion scan progression tracking, objective responses, and PSA progressions.
ADTTE	BASIC DATA STRUCTURE	One record per subject per time-to-event parameter	19	Survival endpoints including Overall Survival (OS), Progression-Free Survival (PFS), TTPSA, and time-to-SAE endpoints.

- ADSL must contain exactly one record per subject and all population, treatment, stratification and key baseline variables needed for analysis.
- BDS datasets must include PARAM/PARAMCD, AVAL/AVALC as applicable, analysis dates/days, visit/window variables and analysis flags.
- OCCDS datasets must retain occurrence identity and derivation flags needed for incidence and episode analyses.
- Define-XML v2.1 with ARM must align to physical datasets and SAP output catalog with no unresolved OIDs, WhereClauses, codelists or ResultDisplay references.
- Every ADaM variable requires origin, predecessor, method and document reference. The current workbook Documents sheet is empty and must be remediated before release.
- SDTMIG version claims must match the packaged tabulation datasets. If SDTMIG 3.4 is declared, the uplifted 3.4 layer must be the packaged source.

16. Programming, independent QC and release gates

Gate	Pass criterion
SAP approval	Signed v4.0 or later approved before production rerun.
Dependency graph	No unexplained orphan, dangling reference, dead production code, specified-not-produced or produced-not-specified item.
Dual programming	SAS and R outputs match on keys, row counts, typed values, labels and hashes without lossy normalization.
Metadata/data concordance	ADaM spec, Define-XML, physical XPT and reviewer guides agree on datasets, variables, labels, types, lengths, codelists and methods.
Conformance	Supported CORE/Pinnacle 21/Define validation run on final delivered package; every issue dispositioned.
Logs	Production and QC logs clean or reviewed with approved exception list.
TFL QC	Every output has shell, SAP section, program, QC result, source data and approved status.
Package integrity	eCTD built atomically; every payload indexed; every leaf exists; hashes match final evidence manifest.
Part 11/process	Validated execution environment, access controls, audit trail and approval

Gate	Pass criterion
	evidence available for regulated claims.

16.1 Dual-programming reconciliation specification

Object	Required comparison
ADaM datasets	Dataset existence, row count, keys, column order, type, length, label, codelist values, typed cell values and record-level hashes.
TFL numeric results	Independent recomputation of numerator, denominator, percentage, event count, median, CI, HR and p-value from analysis datasets.
Figures	Independent verification of input data, at-risk counts, axis labels, treatment labels, footnotes and synthetic/non-confirmatory annotation.
Listings	Record-level source traceability; no fabricated records; all dates and subject IDs sourced from data.
Logs	Automated scan plus human disposition of ERROR, WARNING, invalid INPUT, uninitialized variables, merge notes and package warnings.
Hash manifest	Every release dataset/output hash equals current physical payload and eCTD/index copy.

16.2 Release decision classes

Class	Meaning
Green	Source, program, QC, metadata, TFL and package evidence all agree; no open critical/major issue affects the object.
Amber	Object is scientifically usable for internal review with documented non-critical limitation; not submission-ready until limitation closed or approved.
Red	Object is blocked due confirmed defect, missing source basis, synthetic data mislabeling, metadata drift or failed validation.
Exploratory	Object is allowed only for hypothesis generation or pipeline demonstration; cannot support confirmatory clinical claims.

17. TFL catalog and shell control

TFL IDs below are the controlled SAP target catalog. Current physical outputs are implementation evidence, not proof of completion. Outputs that are not produced must either be implemented and QC'd or removed by approved amendment. Outputs produced outside this catalog must be added to the SAP or removed from release.

ID	Title	Dataset(s)	Population	Status
T-11-1	OS: Median, Events/N, HR, 95%CI, log-rank p	ADTTE OS	ITT	Current physical output exists; requires v4 shell/QC status check.
T-11-2	PFS: Median, Events/N, HR, 95%CI, log-rank p	ADTTE PFS	ITT	Current physical output exists; requires v4 shell/QC status check.
T-11-3	PSA Response Rate (>=50% confirmed >=3 wks; PSA>=20 subset)	ADLB+ADSL	ITT PSA>=20	Current physical output exists; requires v4 shell/QC status check.
T-11-4	ORR per RECIST v1.0 (measurable subset)	ADRS+ADSL	ITT Meas	Current physical output exists; requires v4 shell/QC status check.
T-11-5	Pain Response Rate (PAINBL='Y' subset)	ADRS+ADSL	PAINBL='Y'	Current physical output exists; requires v4 shell/QC status check.
T-11-6	Time to Tumour Progression: Median (IQR), HR, p	ADTTE TTUMOR	ITT	Current physical output exists; requires v4 shell/QC status check.
T-11-7	Time to PSA Progression: Median (IQR), HR, p	ADTTE TTPSA	ITT	Current physical output exists; requires v4 shell/QC status check.
T-11-8	Time to Pain Progression: Median (IQR), HR, p	ADTTE TTPAIN	ITT	Current physical output exists; requires v4 shell/QC status check.
T-12-1	OS Sensitivity: RMST at tau=12 months	ADTTE OS	ITT	Target output; implementation status requires reconciliation.
T-12-2	OS Sensitivity: Landmark at 6 and 12 months	ADTTE OS	ITT	Target output; implementation status requires reconciliation.
T-13-1	Exposure: Duration, Cycles, Cumulative Dose, RDI	ADEX	Safety	Planned; not currently produced as a controlled output.
T-13-2	Dose Modifications: Reductions and Delays by Arm	ADEX	Safety	Planned; not currently produced as a controlled output.
T-13-3	Discontinuation Reasons by Arm	ADEX+ADSL	ITT	Planned; not currently produced as a controlled output.
T-14-1	All TEAEs by SOC and PT (n, %)	ADAE	Safety N=371	Planned; not currently produced

ID	Title	Dataset(s)	Population	Status
				as a controlled output.
T-14-2	Grade >=3 TEAEs by SOC and PT (n, %)	ADAE	Safety	Planned; not currently produced as a controlled output.
T-14-3	Serious TEAEs (AESER='Y')	ADAE	Safety	Planned; not currently produced as a controlled output.
T-14-4	TEAEs Leading to Dose Modification or Discontinuation	ADAE	Safety	Planned; not currently produced as a controlled output.
T-14-5	Deaths <=30 Days of Last Dose - cause (reproduces Table 5)	ADAE+ADSL	Safety	Planned; not currently produced as a controlled output.
T-14-6	Full Deaths Table (reproduces Table 5 complete)	ADSL+ADAE	Safety	Planned; not currently produced as a controlled output.
T-15-1	G-CSF Use by Arm, Cycle, Type (Prophylactic / Therapeutic)	ADCM	Safety	Planned; not currently produced as a controlled output.
T-16-1	Haematology Values by Visit and Arm	ADLB	Safety	Planned; not currently produced as a controlled output.
T-16-2	Worst CTCAE v3.0 Grade per Haematology Parameter	ADLB	Safety	Planned; not currently produced as a controlled output.
T-17-1	RDI by Arm and Cycle (Optimus)	ADEX	CbzP Safety	Current physical output exists; requires v4 shell/QC status check.
T-17-2	ANC Nadir by Cycle x G-CSF Status (Optimus)	ADLB+ADCM	CbzP Safety	Current physical output exists; requires v4 shell/QC status check.
T-17-3	ANC Recovery Latency vs. Delay Status (Optimus)	ADLB+ADEX	CbzP Safety	Current physical output exists; requires v4 shell/QC status check.
T-17-4	Benefit-Risk by RDI Tertile (Optimus)	ADEX+ADTTE+ADAE	CbzP	Current physical output exists; requires v4 shell/QC status check.
F-11-1	KM Curve - OS - reproduce Lancet Figure 2A	ADTTE OS	ITT	Current physical output exists; requires v4 shell/QC status check.
F-11-2	KM Curve - PFS - reproduce Lancet Figure 3	ADTTE PFS	ITT	Current physical output exists; requires v4 shell/QC status check.
F-12-1	Forest Plot - OS Subgroups - reproduce Lancet Figure 2B	ADTTE+ADSL	ITT	Current physical output exists; requires v4 shell/QC status check.
F-12-2	Forest Plot - PFS Subgroups	ADTTE+ADSL	ITT	Planned; not currently produced as a controlled output.
F-17-1	Scatter: RDI vs ANC Nadir (Optimus E-R)	ADEX+ADLB+ADCM	CbzP	Current physical output exists; requires v4 shell/QC status check.

17.1 Shell requirements for every TFL

- Shell must state population, denominator, treatment columns, statistic rows, sorting rules, precision, missing/NE handling and footnotes.
- Shell must identify whether the output is confirmatory, supportive, safety, exploratory or synthetic-comparator demonstration.
- Shell must map to SAP section and ADaM datasets/variables before programming begins.
- A produced output without shell approval is not release-ready even if code runs and numbers look plausible.
- Any output not in the controlled catalog must be removed or added by SAP amendment before package release.

18. Known limitations and remediation controls

The limitations below are not cosmetic. They control whether the project can make clinical, regulatory or submission-readiness claims.

Finding	Severity	Category	Required closure
F-001	Critical	eCTD integrity	Delete the sequence; rebuild full or preview atomically in a new directory; fail on any unindexed payload or missing leaf; verify after final copy.
F-002	Critical	SDTM metadata/data drift	Package only the uplifted 3.4 layer; define all delivered domains/variables; rerun full-package CORE/P21 and metadata/data comparison.
F-003	Critical	Scientific validity / synthetic data	Do not represent illustrative reconstructed/simulated comparator results as submission analyses. Obtain the complete clinical data and rerun, or clearly segregate this repository as a non-regulatory demonstration.
F-004	Critical	False clinical listing	Remove the placeholder from m5/eCTD. Generate the listing from ADSL/DS/EX with independent QC and approved shells.
F-005	Critical	Submission placeholders	Replace application/submission metadata with assigned values and

Finding	Severity	Category	Required closure
			provide the actual annotated CRF with page-level origin links before any submission build.

- Major findings F-006 through F-025 remain release-blocking unless closed or formally risk-accepted by the appropriate owner.
- Synthetic comparator analyses must remain labeled non-confirmatory until complete authoritative CbzP patient-level data are available and analyzed under an approved SAP.
- No reviewer guide may claim clean traceability until generated from reconciled metadata and final program manifests.

19. References

Reference	URL / repo path
FDA Study Data Technical Conformance Guide, June 2026	https://www.fda.gov/regulatory-information/search-fda-guidance-documents/study-data-technical-conformance-guide-technical-specifications-document
FDA Study Data Standards Resources	https://www.fda.gov/industry/fda-data-standards-advisory-board/study-data-standards-resources
ICH/FDA E9(R1) estimands and sensitivity analysis	https://www.fda.gov/regulatory-information/search-fda-guidance-documents/e9r1-statistical-principles-clinical-trials-addendum-estimands-and-sensitivity-analysis-clinical
FDA Project Optimus	https://www.fda.gov/about-fda/oncology-center-excellence/project-optimus
FDA oncology dosage optimization guidance	https://www.fda.gov/regulatory-information/search-fda-guidance-documents/optimizing-dosage-human-prescription-drugs-and-biological-products-treatment-oncologic-diseases
CDISC ADaMIG v1.3	https://www.cdisc.org/standards/foundational/adam/adamig-v1-3
CDISC ADaM v2.1	https://www.cdisc.org/standards/foundational/adam/adam-v2-1
CDISC SDTMIG v3.4	https://www.cdisc.org/standards/foundational/sdtmig/sdtmig-v3-4
CDISC Define-XML v2.1	https://www.cdisc.org/standards/data-exchange/define-xml/define-xml-v2-1
CDISC Controlled Terminology release 2026-03-27	https://www.cdisc.org/standards/terminology/controlled-terminology
21 CFR Part 11	https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-11
TROPIC protocol	01_raw_source/Sanofi Study Protocol Tropic.pdf
TROPIC publication	01_raw_source/reference_literature/de_bono_lancet_2010.pdf

Appendix A. Source register

Source	File / location	SAP use	Reliability / caveat
Protocol Amendment 5, 21-Jul-2008	01_raw_source/Sanofi Study Protocol Tropic.pdf	Objectives, treatment arms, randomization, strata, populations, endpoint definitions, sample size, CTCAE grading, assessment schedule.	High for intended trial design; source is sponsor protocol PDF available in repo.
de Bono et al. Lancet 2010 corrected publication	01_raw_source/reference_literature/de_bono_lancet_2010.pdf	Final cut-off, published ITT/safety counts, primary/secondary results, safety tables, references to CTCAE v3.0 and MedDRA v12.0.	High for public reconciliation targets; not a substitute for original locked analysis datasets.
Sanofi CRF TROPIC	01_raw_source/Sanofi CRF Tropic.pdf	CRF capture context for eligibility, RECIST, labs, pain, AE, exposure, concomitant medication and follow-up forms.	Medium-high for collection design; not sufficient alone for derivation algorithms.
PDS / Sanofi public SDTM release	01_raw_source/real_sdtm/*.sas7bdat	Real patient-level MP arm data in the current repo pipeline.	High for MP arm analyses; incomplete for original two-arm confirmatory inference.
ADaM specification workbook	00_specifications/ADaM_spec.xlsx	ADaM datasets, variables, codelists, methods, value-level metadata and WhereClauses.	Implementation baseline; currently requires traceability completion before release authority.
Audit evidence pack	audit/AUDIT_REPORT.md and audit/findings_register.csv	Critical and Major defects that must be closed or risk-accepted before submission-readiness claims.	High for remediation control; not a clinical source document.
Reconstruction programs and digitized KM inputs	01_raw_source/reconstruct_cbzp_guyot.R; 01_raw_source/reconstruct_cbzp_arm.R; 01_raw_source/guyot_digitised/*	Synthetic/reconstructed CbzP data generation for demonstration.	Valid only for explicitly non-confirmatory demonstration unless formally justified and approved.

Appendix B. Endpoint algorithm details

Endpoint	ADaM target	Key sources	Algorithm control
OS	ADTTE.OS	RANDDT, DTHDT, LSTALVDT, cut-off	Event if death observed; otherwise censor at min(last known alive, cut-off).
PFS	ADTTE.PFS	ADRS PSA/tumour/pain components, death, ADCM.NACTDT	Event date is first valid component; censor before new therapy or at randomization/no post-baseline assessment.
TTPSA	ADTTE.TTPSA	PSA values in ADLB/ADRS	Apply responder/non-responder progression rules and confirmation timing.
TTUMOR	ADTTE.TTUMOR	RECIST ADRS/LS	Event on radiographic PD/new lesion; censor at last evaluable tumour assessment.
TTPAIN	ADTTE.TTPAIN	PN/pain diary and palliative RT CM/PR	Event on pain progression criteria; apply diary evaluability and confirmation rules.
PSA response	ADRS.PSARESP	Baseline/post-baseline PSA	Confirmed >=50% reduction from baseline, repeat >=3 weeks later; population PSA baseline >=20 ug/L.
ORR	ADRS.OBJRESP	RECIST target, non-target, new lesions	Confirmed CR/PR in measurable disease population.
TEAE	ADAE	AE, EX, death, severity, relationship	Treatment emergent if onset/worsening on/after first dose and within safety window.
Lab shift	ADLB	LB value/grade, baseline flag, ANL01FL	One baseline and one worst post-baseline per subject/parameter before shift table.

Appendix C. ADaM dataset and metadata requirements

Dataset	Vars	Mandatory	Derived	Codelisted	Representative variables
ADAE	29	6	1	5	STUDYID, USUBJID, AEDECOD, AEBODSYS, AEHLT, AESEV, ATOXGR, AESER, AEREL, ASTDT …
ADCM	15	5	0	5	STUDYID, USUBJID, CMDECOD, CMCAT, CMINDC, ASTDT, AENDT, CMTRT, ASTDY, GCSFFL …

Dataset	Vars	Mandatory	Derived	Codelisted	Representative variables
ADEX	14	10	4	3	STUDYID, USUBJID, SUBJID, TRT01P, TRT01PN, TRTSDT, PARAMCD, PARAM, PARCAT1, AVAL …
ADLB	27	11	9	4	STUDYID, USUBJID, SUBJID, TRT01P, TRTSDT, PARAMCD, PARAM, PARAMN, PARCAT1, AVAL …
ADRS	13	9	2	3	STUDYID, USUBJID, SUBJID, TRT01P, TRTSDT, PARAMCD, PARAM, AVALC, ADT, ADY …
ADSL	42	20	16	20	STUDYID, USUBJID, SUBJID, SITEID, AGE, AGEGR1, AGEGR1N, RACE, ETHNIC, SEX …
ADTTE	19	14	7	6	STUDYID, USUBJID, SUBJID, SITEID, TRT01P, TRT01PN, ITTFL, SAFFL, PARAMCD, PARAM …

Metadata remediation rule: the Documents sheet, Predecessor metadata and Method document references must be populated before Define-XML is considered release-ready. The current workbook is a starting point, not final authority.

Appendix D. Full TFL catalog implementation notes

ID	SAP section	Implementation note
T-11-1	Section 4	Current physical output exists; requires v4 shell/QC status check.
T-11-2	Section 5.1	Current physical output exists; requires v4 shell/QC status check.
T-11-3	Section 5.2	Current physical output exists; requires v4 shell/QC status check.
T-11-4	Section 5.3	Current physical output exists; requires v4 shell/QC status check.
T-11-5	Section 6.4	Current physical output exists; requires v4 shell/QC status check.
T-11-6	Section 5.3	Current physical output exists; requires v4 shell/QC status check.
T-11-7	Section 5.2	Current physical output exists; requires v4 shell/QC status check.
T-11-8	Section 6.5	Current physical output exists; requires v4 shell/QC status check.
T-12-1	Section 4.7	Target output; implementation status requires reconciliation.
T-12-2	Section 4.7	Target output; implementation status requires reconciliation.
T-13-1	Section 7.8	Planned; not currently produced as a controlled output.
T-13-2	Section 7.8	Planned; not currently produced as a controlled output.
T-13-3	Section 7.4	Planned; not currently produced as a controlled output.
T-14-1	Section 7	Planned; not currently produced as a controlled output.
T-14-2	Section 7	Planned; not currently produced as a controlled output.
T-14-3	Section 7	Planned; not currently produced as a controlled output.
T-14-4	Section 7	Planned; not currently produced as a controlled output.
T-14-5	Section 7.5.1	Planned; not currently produced as a controlled output.
T-14-6	Section 7.5.1	Planned; not currently produced as a controlled output.
T-15-1	Section 7.6	Planned; not currently produced as a controlled output.
T-16-1	Section 7.5	Planned; not currently produced as a controlled output.
T-16-2	Section 7.5	Planned; not currently produced as a controlled output.
T-17-1	Section 10	Current physical output exists; requires v4 shell/QC status check.
T-17-2	Section 10	Current physical output exists; requires v4 shell/QC status check.
T-17-3	Section 10	Current physical output exists; requires v4 shell/QC status check.
T-17-4	Section 10	Current physical output exists; requires v4 shell/QC status check.
F-11-1	Section 4	Current physical output exists; requires v4 shell/QC status check.

ID	SAP section	Implementation note
F-11-2	Section 5.1	Current physical output exists; requires v4 shell/QC status check.
F-12-1	Section 8	Current physical output exists; requires v4 shell/QC status check.
F-12-2	Section 8	Planned; not currently produced as a controlled output.
F-17-1	Section 10	Current physical output exists; requires v4 shell/QC status check.

Appendix E. Published validation targets

Statistic	Published Value	Tolerance	ADaM Source
Median OS - CbzP	15.1 months (95% CI 14.1-16.3)	+/-0.3 mo	ADTTE PARAMCD='OS'
Median OS - MP	12.7 months (95% CI 11.6-13.7)	+/-0.3 mo	ADTTE PARAMCD='OS'
OS HR (CbzP vs. MP)	0.70 (95% CI 0.59-0.83)	+/-0.02	ADTTE + Cox PH
OS log-rank p-value	<0.0001	p<0.001 acceptable	ADTTE
OS deaths - CbzP (ITT)	234	Exact	ADTTE CNSR=0
OS deaths - MP (ITT)	279	Exact	ADTTE CNSR=0
OS deaths total (ITT)	513	Exact	ADTTE CNSR=0 sum
Median follow-up (combined)	12.8 months (IQR 7.8-16.9)	+/-0.5 mo	ADSL derived
Median PFS - CbzP	2.8 months (95% CI 2.4-3.0)	+/-0.2 mo	ADTTE PARAMCD='PFS'
Median PFS - MP	1.4 months (95% CI 1.4-1.7)	+/-0.2 mo	ADTTE PARAMCD='PFS'
PFS HR	0.74 (95% CI 0.64-0.86)	+/-0.02	ADTTE + Cox PH
PFS p-value	<0.0001	-	-
Median TTPSA - CbzP	6.4 months (IQR 2.2-10.1)	+/-0.3 mo	ADTTE PARAMCD='TTPSA'
Median TTPSA - MP	3.1 months (IQR 0.9-9.1)	+/-0.3 mo	ADTTE PARAMCD='TTPSA'
TTPSA HR	0.75 (0.63-0.90)	+/-0.02	ADTTE + Cox PH
TTPSA p-value	0.001	-	-
Median TTUMOR - CbzP	8.8 months (IQR 3.9-12.0)	+/-0.3 mo	ADTTE PARAMCD='TTUMOR'
Median TTUMOR - MP	5.4 months (IQR 2.3-10.0)	+/-0.3 mo	ADTTE PARAMCD='TTUMOR'
TTUMOR HR	0.61 (0.49-0.76)	+/-0.02	ADTTE + Cox PH
TTUMOR p-value	<0.0001	-	-
Median TTPAIN - CbzP	11.1 months (IQR 2.9-NR)	+/-0.5 mo	ADTTE PARAMCD='TTPAIN'
Median TTPAIN - MP	Not reached	N/A	ADTTE PARAMCD='TTPAIN'
TTPAIN HR	0.91 (0.69-1.19)	+/-0.02	ADTTE + Cox PH
TTPAIN p-value	0.52 (not significant)	-	-
PSA response rate - CbzP	39.2% (33.9-44.5)	+/-1%	ADLB PARAMCD='PSAD50'
PSA response rate - MP	17.8% (13.7-22.0)	+/-1%	ADLB PARAMCD='PSAD50'
PSA response p-value	0.0002	-	-
ORR (RECIST) - CbzP	14.4% (9.6-19.3)	+/-1%	ADRS PARAMCD='OBJRESP'
ORR (RECIST) - MP	4.4% (1.6-7.2)	+/-1%	ADRS PARAMCD='OBJRESP'
ORR p-value	0.0005	-	-
Pain response rate - CbzP	9.2% (4.9-13.5)	+/-1%	ADRS PARAMCD='PAINRESP'
Pain response rate - MP	7.7% (3.7-11.8)	+/-1%	ADRS PARAMCD='PAINRESP'
Pain response p-value	0.63 (not significant)	-	-
Grade >=3 neutropenia - CbzP	82% (303/371)	+/-2%	ADLB (lab-based)
Grade >=3 neutropenia - MP	58% (215/371)	+/-2%	ADLB (lab-based)
Febrile neutropenia G>=3 - CbzP	8% (28/371)	+/-1%	ADAE
Febrile neutropenia G>=3 - MP	1% (5/371)	+/-1%	ADAE
Grade >=3 diarrhoea - CbzP	6% (23/371)	+/-1%	ADAE
Any grade peripheral neuropathy - CbzP	14% (52/371)	+/-1%	ADAE
Total deaths in study - CbzP (safety)	227 (61%)	Exact	ADSL + Table 5
Total deaths in study - MP (safety)	275 (74%)	Exact	ADSL + Table 5
Deaths <=30d - CbzP (safety)	18 (5%)	Exact	ADAE
Deaths <=30d - MP (safety)	9 (2%)	Exact	ADAE
Median cycles - CbzP	6 (IQR 3-10)	Exact	ADEX PARAMCD='NCYCLE'
Median cycles - MP	4 (IQR 2-7)	Exact	ADEX PARAMCD='NCYCLE'
Median RDI - CbzP	96.1% (IQR 90.1-98.9)	+/-0.5%	ADEX PARAMCD='RDI'
Median RDI - MP	97.3% (IQR 92.0-99.3)	+/-0.5%	ADEX PARAMCD='RDI'
Total cycles - CbzP	2,251	Exact	ADEX cycle count
Total cycles - MP	1,736	Exact	ADEX cycle count
Safety population N - each arm	371 per arm	Exact	ADSL SAFFL='Y'
ITT population N - CbzP	378	Exact	ADSL ITTFL='Y'
ITT population N - MP	377	Exact	ADSL ITTFL='Y'

Appendix F. Safety, death and exposure publication targets

F.1 Discontinuation targets

Reason for Discontinuation	MP n=377 (%)	CbzP n=378 (%)
Discontinued study treatment (any reason)	325 (86%)	266 (70%)
- Disease progression	267 (71%)	180 (48%)
- Adverse event	32 (8%)	67 (18%)
- Non-compliance with protocol	0	1 (<1%)
- Lost to follow-up	2 (1%)	0
- Patient request	17 (5%)	8 (2%)
- Other	7 (2%)	10 (3%)
Completed planned 10 cycles	46 (12%)	105 (28%)

F.2 Adverse event targets

Adverse Event	MP Any	MP G>=3	CbzP Any	CbzP G>=3
HAEMATOLOGICAL (lab-based)				
Neutropenia	88% (325)	58% (215)	94% (347)	82% (303)
Febrile neutropenia	-	1% (5)	-	8% (28)
Leukopenia	92% (343)	42% (157)	96% (355)	68% (253)
Anaemia	81% (302)	5% (18)	97% (361)	11% (39)
Thrombocytopenia	43% (160)	2% (6)	47% (176)	4% (15)
NON-HAEMATOLOGICAL				
Diarrhoea	11% (39)	<1% (1)	47% (173)	6% (23)
Fatigue	27% (102)	3% (11)	37% (136)	5% (18)
Nausea	23% (85)	<1% (1)	34% (127)	2% (7)
Constipation	15% (57)	1% (2)	20% (76)	1% (4)
Vomiting	10% (38)	0	23% (84)	2% (7)
Asthenia	12% (46)	2% (9)	20% (76)	5% (17)
Back pain	12% (45)	3% (11)	16% (60)	4% (14)
Arthralgia	8% (31)	1% (4)	11% (39)	1% (4)
Pain in extremity	7% (27)	1% (4)	8% (30)	2% (6)
Haematuria	4% (14)	1% (2)	17% (62)	2% (7)
Abdominal pain	4% (13)	0	12% (43)	2% (7)
Dyspnoea	5% (17)	1% (3)	12% (44)	1% (5)
Pyrexia	6% (23)	<1% (1)	12% (45)	1% (4)
Urinary tract infection	3% (11)	1% (3)	7% (27)	1% (4)
Pain	5% (18)	2% (7)	5% (20)	1% (4)
Bone pain	5% (19)	2% (9)	5% (19)	1% (3)
Peripheral neuropathy	3% (12)	1% (3)	14% (52)	1% (3)
Peripheral oedema	9% (34)	-	9% (34)	-

F.3 Death targets

Death Parameter	MP (n=371 safety)	CbzP (n=371 safety)
Total deaths during study (safety population)	275 (74%)	227 (61%)
Deaths <=30 days after last dose	9 (2%)	18 (5%)
- Disease progression	6 (2%)*	0
- Neutropenia and clinical consequences / sepsis	1 (<1%)	7 (2%)
- Cardiac	0	5 (1%)
- Renal failure	0	3 (1%)
- Dyspnoea (investigator: disease progression)*	1 (<1%)	0
- Dehydration / electrolyte imbalance	0	1 (<1%)
- Cerebral haemorrhage	0	1 (<1%)
- Unknown cause	0	1 (<1%)
- Motor vehicle accident	1 (<1%)	0
Deaths >30 days after last dose	266 (72%)	209 (56%)

F.4 Exposure targets

Exposure Parameter	MP	CbzP
Median cycles (IQR)	4 (IQR 2-7)	6 (IQR 3-10)
Patients completing all 10 cycles	46 (12%)	105 (28%)
Patients receiving >90% planned dose intensity	301 (81%)	282 (76%)

Exposure Parameter	MP	CbzP
Median RDI % (IQR)	97.3% (92.0-99.3)	96.1% (90.1-98.9)
Patients with >=1 dose reduction	15 (4%)	45 (12%)
Dose reduction cycles / total cycles	88 / 1,736 (5%)	221 / 2,251 (10%)
Patients with any treatment delay	56 (15%)	104 (28%)
Delay cycles <=9 days / total	110 (6%)	157 (7%)
Delay cycles >9 days / total	28 (2%)	51 (2%)
Total treatment cycles	1,736	2,251

Appendix G. Subgroup target register

Subgroup	N*	OS HR (95% CI)	ADSL Variable
All randomised	755	0.70 (0.59-0.83)	Full ITT
ECOG PS 0-1	694	0.68 (0.57-0.82)	ECOGBLN IN (0,1)
ECOG PS 2	61	0.81 (0.48-1.38)	ECOGBLN=2
Non-measurable disease	350	0.72 (0.55-0.93)	MEASDISFL='N'
Measurable disease	405	0.68 (0.54-0.85)	MEASDISFL='Y'
Prior chemo regimens: 1	528	0.67 (0.55-0.83)	NPRCHEMO=1
Prior chemo regimens: >=2	227	0.75 (0.55-1.02)	NPRCHEMO GE 2
Age <65 years	295	0.81 (0.61-1.08)	AGEGR1='<65'
Age >=65 years	460	0.62 (0.50-0.78)	AGEGR1='>=65'
No pain at baseline*a	374	0.55 (0.42-0.72)	PAINBL='N'
Pain at baseline*a	342	0.77 (0.61-0.98)	PAINBL='Y'
No rising PSA at baseline*b	159	0.88 (0.61-1.26)	RISEPSAFL='N'
Rising PSA at baseline*b	583	0.65 (0.53-0.80)	RISEPSAFL='Y'
Prior docetaxel <225 mg/m^2*c	59	0.96 (0.49-1.86)	TOTDOC_GRP='<225'
Prior docetaxel >=225-<450 mg/m^2*c	206	0.60 (0.43-0.84)	TOTDOC_GRP='225<450'
Prior docetaxel >=450-<675 mg/m^2*c	217	0.83 (0.60-1.16)	TOTDOC_GRP='450<675'
Prior docetaxel >=675-<900 mg/m^2*c	131	0.73 (0.48-1.10)	TOTDOC_GRP='675<900'
Prior docetaxel >=900 mg/m^2*c	134	0.51 (0.33-0.79)	TOTDOC_GRP='>=900'
Progression during docetaxel	219	0.65 (0.47-0.90)	DOCPROG='DURING'
Progression <3 months after last dose	339	0.70 (0.55-0.91)	DOCPROG='<3MO'
Progression >=3 months after last dose	192	0.75 (0.51-1.11)	DOCPROG='>=3MO'

Appendix H. Visit-window and protocol-correction registers

H.1 ADLB visit-window rules

AVISITN	AVISIT Label	ADY Low	ADY High	Target Day	Note
0	Baseline	<=0	<=0	Last pre-dose	ABLFL='Y'; ATOXGR from LBTOXGR
1	Cycle 1 Day 1 Pre-dose	1	3	1	Scheduled CBC; go/no-go gate C1
2	Cycle 1 Day 8	4	13	8	Primary C1 nadir; G-CSF prohibited
3	Cycle 1 Day 15	14	17	15	C1 recovery check
4	Cycle 2 Day 1 Pre-dose	18	24	22	Scheduled CBC; gate for C2
5	Cycle 2 Day 8	25	34	29	C2 nadir
6	Cycle 2 Day 15	35	38	36	C2 recovery check
7	Cycle 3 Day 1 Pre-dose	39	45	43	Gate for C3
8	Cycle 3 Day 8	46	55	50	C3 nadir
99	Unscheduled	All others	-	-	ANL01FL='N'

H.2 Protocol versus publication correction register

Item	Protocol Value	Paper Value - USE THIS	Section	v3 Error #
Data cut-off	Protocol data lock reference ~Mar 2010	25-Sep-2009 (paper p.1148)	Section 1.3	-
Final OS alpha	0.0476 (O'Brien-Fleming)	0.0452 (paper p.1151)	Section 4.5	-
OS IA trigger	307 OS deaths	365 actual; alpha=0.016 (paper p.1151)	Section 4.5	-
Safety N - CbzP arm	378 (all randomised)	371 (378-7); paper Figure 1	Section 3.2	Error 1
Safety N - MP arm	377 (all randomised)	371 (377-6); paper Figure 1	Section 3.2	Error 1
PSA progression - responders	>=50% over nadir AND >=5ug/L	>=50% over nadir ONLY (paper p.1150)	Section 5.2.2	Error 2
Pain progression Criterion 1	FROM NADIR	From REFERENCE VALUE = baseline (paper p.1150)	Section 6.5	Error 3
G-CSF trigger (post-C1)	After febrile neutropenia	After neutropenia >=7d OR	Section 7.6	-

Item	Protocol Value	Paper Value - USE THIS	Section	v3 Error #
	(mandatory)	fever/infection (discretionary) (paper p.1150)		
PSA progression confirmation	>=1 week (protocol Section 9.1.2.1)	Not stated in paper; protocol rule applied conservatively	Section 5.2.2	-
AS pain response criterion	>=50% (protocol)	'more than 50%' (paper body) vs '50% or more' (Table 3 fn); use >=50%	Section 6.4	-
Total cycles	Not pre-specified	1,736 MP; 2,251 CbzP (Table 2)	Section 7.8	-
SAS version	Not stated	SAS 9.1.3 (paper Methods)	Section 13	-

Appendix I. Full ADaM variable specification summary

Dataset	Variable	Label	Type	Req	Codelist	Origin
ADSL	STUDYID	Study Identifier	text	Yes		Protocol
ADSL	USUBJID	Unique Subject Identifier	text	Yes		Collected
ADSL	SUBJID	Subject Identifier for the Study	text	Yes		Collected
ADSL	SITEID	Study Site Identifier	text	Yes		Collected
ADSL	AGE	Age	float	Yes		Collected
ADSL	AGEGR1	Pooled Age Group 1	text	Yes	CL.AGEGR	Collected
ADSL	AGEGR1N	Pooled Age Group 1 (N)	float	Yes		Collected
ADSL	RACE	Race	text	Yes		Collected
ADSL	ETHNIC	Ethnicity	text	No		Collected
ADSL	SEX	Sex	text	Yes	CL.SEX	Collected
ADSL	TRT01P	Planned Treatment for Period 01	text	Yes	CL.TRT	Assigned
ADSL	TRT01PN	Planned Treatment for Period 01 (N)	float	Yes	CL.TRT01PN	Assigned
ADSL	TRT01A	Actual Treatment for Period 01	text	Yes	CL.TRT	Assigned
ADSL	TRT01AN	Actual Treatment for Period 01 (N)	float	Yes	CL.TRT01PN	Assigned
ADSL	RANDDT	Randomization Date	date	Yes		Collected
ADSL	TRTSDT	Date of First Exposure to Treatment	date	Yes		Derived
ADSL	TRTEDT	Date of Last Exposure to Treatment	date	No		Derived
ADSL	TRTDURD	Total Treatment Duration (Days)	float	No		Derived
ADSL	ITTFL	Intent-to-Treat Population Flag	text	Yes	CL.NY	Derived
ADSL	SAFFL	Safety Population Flag	text	Yes	CL.NY	Derived
ADSL	PPROTFL	Per-Protocol Population Flag	text	Yes	CL.NY	Collected
ADSL	DTHFL	Subject Death Flag	text	Yes	CL.NY	Collected
ADSL	DTHDT	Date of Death	date	No		Derived
ADSL	DTHCAUS	Primary Cause of Death	text	No		Collected
ADSL	LSTALVDT	Date Last Known Alive	date	Yes		Derived
ADSL	ECOGBL	Baseline ECOG Performance Status	float	No		Derived
ADSL	MEASDISF	Baseline Measurable Disease Flag	text	No	CL.NY	Derived
ADSL	VISCFL	Baseline Visceral Metastasis Flag	text	No	CL.NY	Collected
ADSL	PAINBL	Baseline Pain Assessment Flag	text	No	CL.NY	Derived
ADSL	PSABL	Baseline PSA (ug/L)	float	No		Collected
ADSL	ALPBL	Baseline Alkaline Phosphatase (U/L)	float	No		Collected
ADSL	ALBBL	Baseline Albumin (g/L)	float	No		Assigned
ADSL	LDHBL	Baseline Lactate Dehydrogenase (U/L)	float	No		Assigned
ADSL	HGBBL	Baseline Hemoglobin (g/dL)	float	No		Collected
ADSL	DOCPROG	Docetaxel Prior Progression Class	text	No		Collected
ADSL	DOCRESP	Docetaxel Prior Response Flag	text	No	CL.NY	Collected
ADSL	ECOGBLIF	Baseline ECOG Imputed Flag	text	No	CL.NY	Derived
ADSL	PSABLIF	Baseline PSA Imputed Flag	text	No	CL.NY	Derived
ADSL	ALPBLIF	Baseline ALP Imputed Flag	text	No	CL.NY	Derived
ADSL	HGBBLIF	Baseline Hemoglobin Imputed Flag	text	No	CL.NY	Derived
ADSL	ALBBLIF	Baseline Albumin Imputed Flag	text	No	CL.NY	Derived
ADSL	LDHBLIF	Baseline LDH Imputed Flag	text	No	CL.NY	Derived

Dataset	Variable	Label	Type	Req	Codelist	Origin
ADEX	STUDYID	Study Identifier	text	Yes		Protocol
ADEX	USUBJID	Unique Subject Identifier	text	Yes		Collected
ADEX	SUBJID	Subject Identifier for the Study	text	Yes		Collected
ADEX	TRT01P	Planned Treatment for Period 01	text	Yes	CL.TRT	Assigned
ADEX	TRT01PN	Planned Treatment for Period 01 (N)	float	Yes	CL.TRT01PN	Assigned
ADEX	TRTSDT	Date of First Exposure to Treatment	date	Yes		Collected
ADEX	PARAMCD	Parameter Code	text	Yes	CL.PARAMCD	Collected
ADEX	PARAM	Parameter	text	Yes		Collected
ADEX	PARCAT1	Parameter Category 1	text	Yes		Collected
ADEX	AVAL	Analysis Value	float	No		Derived
ADEX	AVALC	Analysis Value (C)	text	No		Derived
ADEX	AVISIT	Analysis Visit	text	Yes		Collected
ADCM	STUDYID	Study Identifier	text	Yes		Protocol
ADCM	USUBJID	Unique Subject Identifier	text	Yes		Collected
ADCM	CMDECOD	Standardized Medication Name	text	Yes		Collected
ADCM	CMCAT	Category for Medication	text	Yes		Collected
ADCM	CMINDC	Indication	text	No		Collected
ADCM	ASTDT	Analysis Start Date	date	No		Collected
ADCM	AENDT	Analysis End Date	date	No		Collected
ADCM	CMTRT	Reported Name of Drug, Med, or Therapy	text	Yes		Collected
ADCM	ASTDY	Analysis Start Day	float	No		Collected
ADCM	GCSFFL	G-CSF Concomitant Med Flag	text	No	CL.NY	Collected
ADCM	GCSFPRFL	G-CSF Prophylactic Med Flag	text	No	CL.NY	Collected
ADCM	NACTFL	New Anti-Cancer Therapy Flag	text	No	CL.NY	Collected
ADCM	NACTDT	First Date of New Anti-Cancer Therapy	date	No		Collected
ADCM	PREDNFL	Prednisone Co-Medication Flag	text	No	CL.NY	Collected
ADCM	TRTEMFL	Treatment Emergent Med Flag	text	No	CL.NY	Assigned
ADAE	STUDYID	Study Identifier	text	Yes		Protocol
ADAE	USUBJID	Unique Subject Identifier	text	Yes		Collected
ADAE	AEDECOD	Dictionary-Derived Term	text	Yes		Collected
ADAE	AEBODSYS	Body System or Organ Class	text	Yes		Collected
ADAE	AEHLT	High Level Term	text	No		Collected
ADAE	AESEV	Severity/Intensity	text	No		Collected
ADAE	ATOXGR	Analysis Toxicity Grade	float	No		Derived
ADAE	AESER	Serious Event	text	No	CL.NY	Collected
ADAE	AEREL	Causality	text	No		Collected
ADAE	ASTDT	Analysis Start Date	date	No		Collected
ADAE	AENDT	Analysis End Date	date	No		Collected
ADAE	ASTDY	Analysis Start Day	float	No		Collected
ADAE	AENDY	Analysis End Day	float	No		Collected
ADAE	AEACN	Action Taken with Study Treatment	text	No		Collected
ADAE	AEOUT	Outcome of Adverse Event	text	No		Collected
ADAE	CQ02NAM	Customized Query 02 Name	text	No		Collected
ADAE	CQ02CD	Customized Query 02 Code	text	No		Collected
ADAE	CQ02SC	Customized Query 02 Source	text	No		Collected
ADAE	CIAESEQ	Custom Episode Sequence Number	float	No		Collected
ADAE	CIAESDT	Custom Episode Start Date	date	No		Collected
ADAE	CIAEEDT	Custom Episode End Date	date	No		Collected
ADAE	CIAEDUR	Custom Episode Duration (Months)	float	No		Collected
ADAE	AEOCCFL	Adverse Event Occurrence Flag	text	No	CL.NY	Collected
ADAE	TRTEMFL	Treatment Emergent AE	text	No	CL.NY	Assigned

Dataset	Variable	Label	Type	Req	Codelist	Origin
		Flag				
ADAE	ADURN	AE Duration (Numeric)	float	No		Collected
ADAE	ADURU	AE Duration Units	text	No		Collected
ADAE	AESEQ	Adverse Event Sequence Number	integer	No		Collected
ADAE	TRTA	Actual Treatment	text	Yes	CL.TRT	Assigned
ADAE	TRTAN	Actual Treatment (N)	float	Yes	CL.TRT01PN	Assigned
ADLB	STUDYID	Study Identifier	text	Yes		Protocol
ADLB	USUBJID	Unique Subject Identifier	text	Yes		Collected
ADLB	SUBJID	Subject Identifier for the Study	text	Yes		Collected
ADLB	TRT01P	Planned Treatment for Period 01	text	Yes	CL.TRT	Assigned
ADLB	TRTSDT	Date of First Exposure to Treatment	date	Yes		Collected
ADLB	PARAMCD	Parameter Code	text	Yes	CL.PARAMCD	Collected
ADLB	PARAM	Parameter	text	Yes		Collected
ADLB	PARAMN	Parameter (N)	float	Yes		Collected
ADLB	PARCAT1	Parameter Category 1	text	Yes		Collected
ADLB	AVAL	Analysis Value	float	No		Derived
ADLB	AVALC	Analysis Value (C)	text	No		Derived
ADLB	LBNRLO	Reference Range Lower Limit in Orig Unit	text	No		Collected
ADLB	LBNRHI	Reference Range Upper Limit in Orig Unit	text	No		Collected
ADLB	LBNRIND	Reference Range Indicator	text	No		Collected
ADLB	AVISIT	Analysis Visit	text	Yes		Collected
ADLB	AVISITN	Analysis Visit (N)	float	Yes		Collected
ADLB	AWDIST	Analysis Window Distance	float	No		Collected
ADLB	ATOXGR	Analysis Toxicity Grade	float	No		Derived
ADLB	BASE	Baseline Value	float	No		Derived
ADLB	BASEC	Result or Finding in Original Units	text	No		Derived
ADLB	BTOXGR	Baseline Toxicity Grade	float	No		Collected
ADLB	CHG	Change from Baseline	float	No		Derived
ADLB	PCHG	Percent Change from Baseline	float	No		Derived
ADLB	ANL01FL	Analysis Record Flag 01	text	No	CL.NY	Collected
ADLB	BASEFL	Baseline Record Flag	text	No	CL.NY	Derived
ADLB	LBDY	Analysis Day	float	No		Collected
ADLB	ADT	Analysis Date	date	No		Derived
ADRS	STUDYID	Study Identifier	text	Yes		Protocol
ADRS	USUBJID	Unique Subject Identifier	text	Yes		Collected
ADRS	SUBJID	Subject Identifier for the Study	text	Yes		Collected
ADRS	TRT01P	Planned Treatment for Period 01	text	Yes	CL.TRT	Assigned
ADRS	TRTSDT	Date of First Exposure to Treatment	date	Yes		Collected
ADRS	PARAMCD	Parameter Code	text	Yes	CL.PARAMCD	Collected
ADRS	PARAM	Parameter	text	Yes		Collected
ADRS	AVALC	Analysis Value (C)	text	No		Derived
ADRS	ADT	Analysis Date	date	Yes		Collected
ADRS	ADY	Analysis Day	float	No		Collected
ADRS	AVISIT	Analysis Visit	text	Yes		Collected
ADRS	ANL01FL	Analysis Record Flag 01	text	No	CL.NY	Collected
ADRS	AVAL	Analysis Value	float	No		Derived
ADTTE	STUDYID	Study Identifier	text	Yes		Protocol
ADTTE	USUBJID	Unique Subject Identifier	text	Yes		Collected
ADTTE	SUBJID	Subject Identifier for the Study	text	Yes		Collected
ADTTE	SITEID	Study Site Identifier	text	Yes		Collected
ADTTE	TRT01P	Planned Treatment for Period 01	text	Yes	CL.TRT	Assigned
ADTTE	TRT01PN	Planned Treatment for Period 01 (N)	float	Yes	CL.TRT01PN	Assigned
ADTTE	ITTFL	Intent-To-Treat Population Flag	text	Yes	CL.NY	Derived
ADTTE	SAFFL	Safety Population Flag	text	Yes	CL.NY	Derived

Dataset	Variable	Label	Type	Req	Codelist	Origin
ADTTE	PARAMCD	Parameter Code	text	Yes	CL.PARAMCD	Collected
ADTTE	PARAM	Parameter	text	Yes		Collected
ADTTE	PARAMN	Parameter (N)	float	No		Derived
ADTTE	PARCAT1	Parameter Category 1	text	No		Derived
ADTTE	STARTDT	Time-to-Event Start Date	date	Yes		Collected
ADTTE	ADT	Analysis Date	date	Yes		Collected
ADTTE	CNSR	Censor Flag	float	Yes	CL.CNSR	Derived
ADTTE	EVNTDESC	Event Description	text	No		Collected
ADTTE	CNSDTDSC	Censoring Description	text	No		Collected
ADTTE	AVAL	Analysis Value (Days)	float	Yes		Derived
ADTTE	AVALU	Analysis Value Unit	text	No		Derived
ADEX	PARAMN	Parameter (N)	float	No		Derived
ADEX	AVISITN	Analysis Visit (N)	float	No		Derived

Appendix J. Critical audit closure checklist

Finding	Closure evidence required
F-001 eCTD integrity	Fresh atomic sequence build; no unindexed payloads; all leaves resolve; hashes match current final outputs.
F-002 SDTM metadata/data drift	Packaged SDTM version matches Define and standards declaration; full dataset/variable reconciliation passes.
F-003 synthetic data validity	All synthetic-comparator outputs relabeled non-confirmatory or replaced by complete authoritative CbzP IPD analysis.
F-004 false listing	Placeholder listing removed; discontinuation listing generated from real source/ADaM with independent QC.
F-005 submission placeholders	Real application metadata and annotated CRF supplied or package generation blocked.